I. Sample

Sample	Nanopore long-read object (noninterleaved)	File size	Illumina short-read object	File size
	CAN1	23 GB	CAN_1_S97	5.7 + 5.8 GB
	CAN2	18 GB	CAN_2_S98	7.3+ 7.5 BG
	CAN3	26 GB	CAN_3_S99	7.2 +6.7 GB
	CAN4	28 GB	CAN_4_S100	6.9 +7 GB
	CAN5	28.5 GB	CAN_5_101	6.0+ 6.1 GB

II. Assembly paths:

Goals in priority:

1.High Quality near or complete MAGs – (proving the presence/lack of denitrifying genes)- the prevalence and role of incomplete denitrifiers (potential cross feeding of denitrification intermediates) and differential gene abundance under various environmental conditions and their roles in N_2O accumulation

2.Presence of ppK genes (P uptake genes) in more novel non-traditional polyphosphate accumulating organisms (PAO)- omics-informed elucidation of diversity and dynamics in common ISC-utilizing denitrifiers and those involved in poly-P accumulation.

3.Potential carbon storage polymer pathways in PAO, GAO, denitrifying groups (formation of PHB, PHV, PH2MV, glycogen and other carbon polymers)

Short	Long	Hybrid
CAN_1_S97	CAN1	CAN_1_S97 + CAN1
FastQC	NanoPlot	
Trimmomatic no phred64 conversion adapter clipping disabled unless FastQC adapter content fails	Filtlong min length: 1000 keep percent: 90 target bases: no limit / full filtered data	Trimmed short reads + filtered long reads

sliding window: 4:20 or 4:30 min length: 50		
FastQC	NanoPlot	
MEGAHIT or metaSPades- (remove the ?G limit?) short-read metagenome assembly/contigs	metaFlye- (remove the 10G limit?) long-read metagenome assembly/contigs	HybridSPAdes -remove memory limit
QUAST	QUAST	QUAST
MaxBin2 or **MetaBAT2?** **Memory limit??** -assemble bins	**MaxBin2** or **MetaBAT2?** **Memory limit??**	**MaxBin2** or **MetaBAT2?** **Memory limit??**
CheckM	CheckM	CheckM
Extract higher quality bins	Polish with short reads?? base correction?? Extract higher quality bins	Extract higher quality bins
RASTtk Annotate Microbial Assembly	RASTtk Annotate Microbial Assembly	Annotate Microbial Assembly with RASTtk
	Taxonomy? Do we have sth for taxnomy	

III. Actual done:

A. Longread assembly

Step 1. Import from Staging Area S

longreads/CAN_1/CAN_1_nanopore.fastq.gz

→ **CAN1**

Step 2. Flye

CAN1

→ CAN1_long_assembly

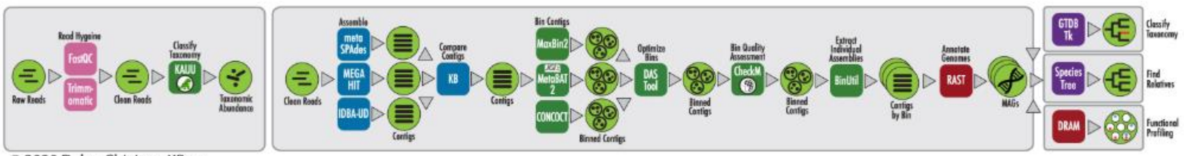
RASTtk metagenome annotation

CAN1_long_assembly

→ CAN1_annotated_metagenome

B. Shortread assembly

https://narrative.kbase.us/narrative/33233



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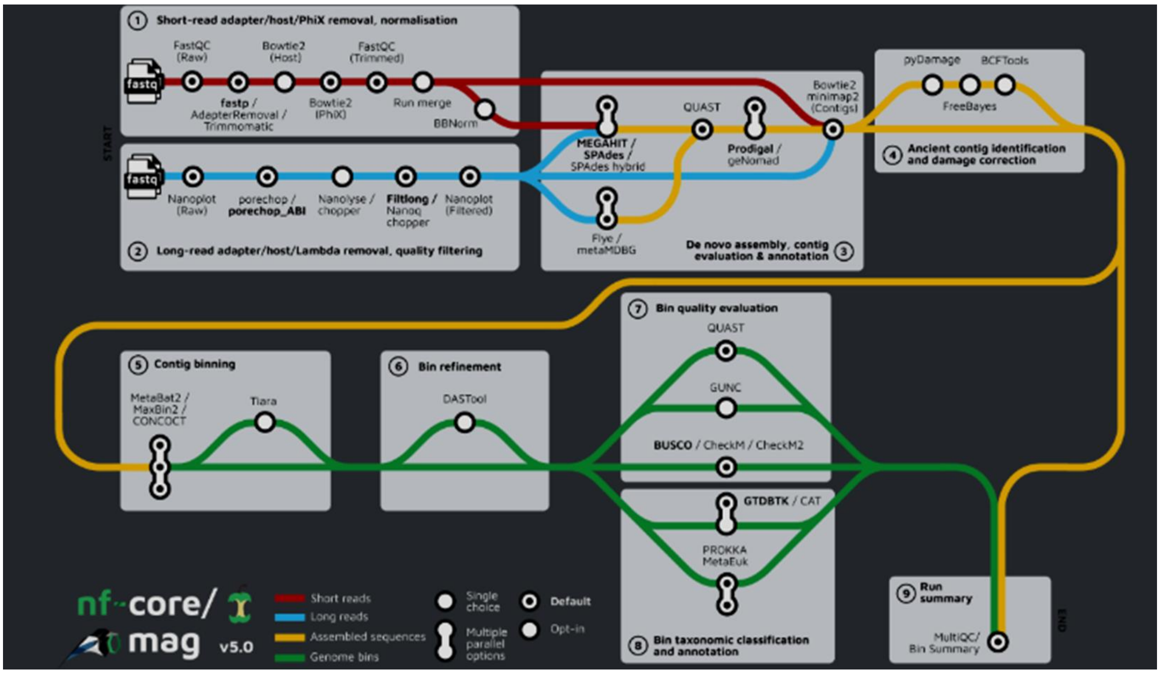
steps	Cell	Function	Settings
	Fast qc	purpose was to find adapter type, get rawread quality	
	Trimmomatic - v0.39:		

C. Hybrid assembly



Illumina short reads ↓ FastQC ↓ Trim Reads with Trimmomatic ↓ FastQC again ↓ filtered/trimmed Illumina read object	Nanopore long reads ↓ NanoPlot ↓ Filter Reads with Filtlong (10GB or 5GB) ↓ NanoPlot again ↓ filtered Nanopore read object

steps	Cell	Function	Settings
	CAN1_filtlong_min1000_keep90 (5gb/10gb)	Filtlong	Filtlong minimum read length = 1000 keep percent = 90 target bases = 5GB or 10 GB
	CAN_1_S97_trimmomatic_SW4Q20_MIN50_noAdapter	Trimmomatic	Translate quality encoding from phred64 to phred33: no Adapter clipping options: disabled Sliding window size: 4 Sliding window minimum quality: 20 Head crop length: 0

			Leading minimum quality: 3 Trailing minimum quality: 3 Minimum read length: 50 Output: CAN_1_S97_trimmomatic_SW4Q20_MIN50_noAdapter
		HybridSPAdes / hybrid assembler	Use 1. CAN_1_S97_trimmomatic_SW4Q20_MIN50_noAdapter_paired ~~2. CAN1_filtlong_min1000_keep90_10G failed~~ 3. AN1_filtlong_min1000_keep90_5G failed

IV. Functional genes to look out for

A. Denitrification genes to check

narG, narH, narI

napA, napB

nirS, nirK

norB, norC

nosZ

B.

acsA	acetate activation
ackA, pta	acetate conversion
prpE	propionate activation
phaA	acetyl-CoA acetyltransferase
phaB	acetoacetyl-CoA reductase
phaC	PHA synthase
phaZ	PHA depolymerase

GlgA, glgB, glgC, glgP, glgX

C.

ppk1	polyphosphate kinase
ppk2	polyphosphate kinase type 2
ppx	exopolyphosphatase
pstS	
pstC	
pstA	
pstB	high-affinity phosphate transporter
pitA	low-affinity phosphate transporter
phoU	

phoB

phoR